CONCEPT 3 — FOUR-BED GARAGE HOUSE

Two storeys · double garage · home office · open kitchen, meals and family to a rear alfresco · four bedrooms + rumpus



Blender render — street view (golden hour)



Blender render — aerial from the south-west

A compact, modern two-storey house with flat roofs behind parapets. The walls are white render downstairs and white vertical cladding upstairs, with a stone pier at the entry and a dark panel garage door. A double garage sits against the blank east party wall. The front door opens to a foyer with the home office beside it and a U-shaped stair behind. A hallway passes the powder room and laundry to the open-plan kitchen, meals and family room across the back, which opens to a covered alfresco facing the river garden. A butler's pantry links the kitchen and the garage. Upstairs, a rumpus room faces the street. The rear bedroom suite has a double walk-in robe and ensuite, and there are three more bedrooms, a bathroom, a powder room and a linen walk-in. A covered deck off Bedroom 4 sits on the garage roof. A 2.8 m garden strip runs down the west side. The DNN land holds only the boundary walls and planting.

THE CONCEPT IN NUMBERS

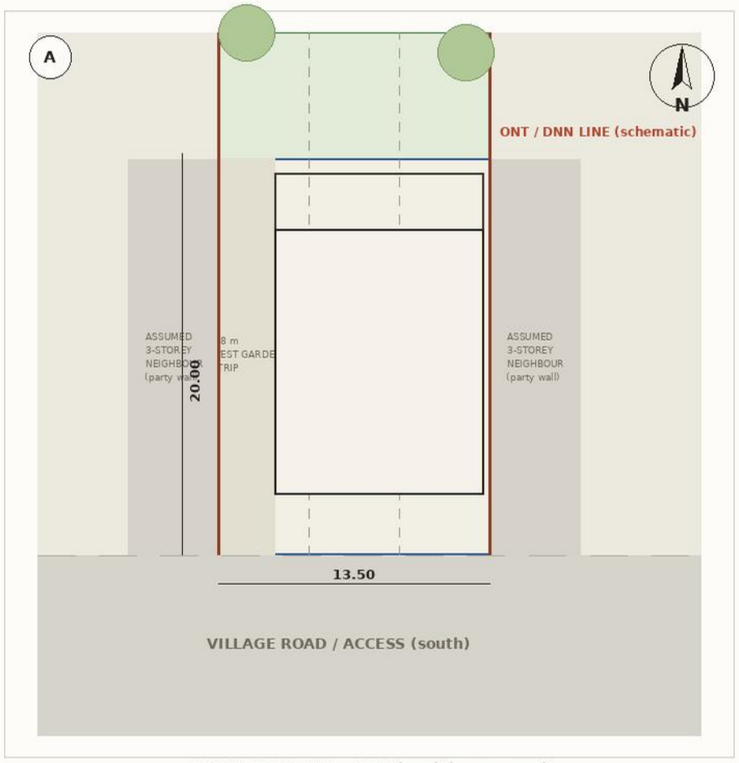
Footprint	137.3 m² (52% of ONT band)
Ground floor	121 m² enclosed
Upper floor	117 m² enclosed + 7 m² balcony
Total enclosed	238 m²
Constraint check	PASS — all structures in ONT; DNN planting only

Walkthrough test: 32/32 rooms reachable

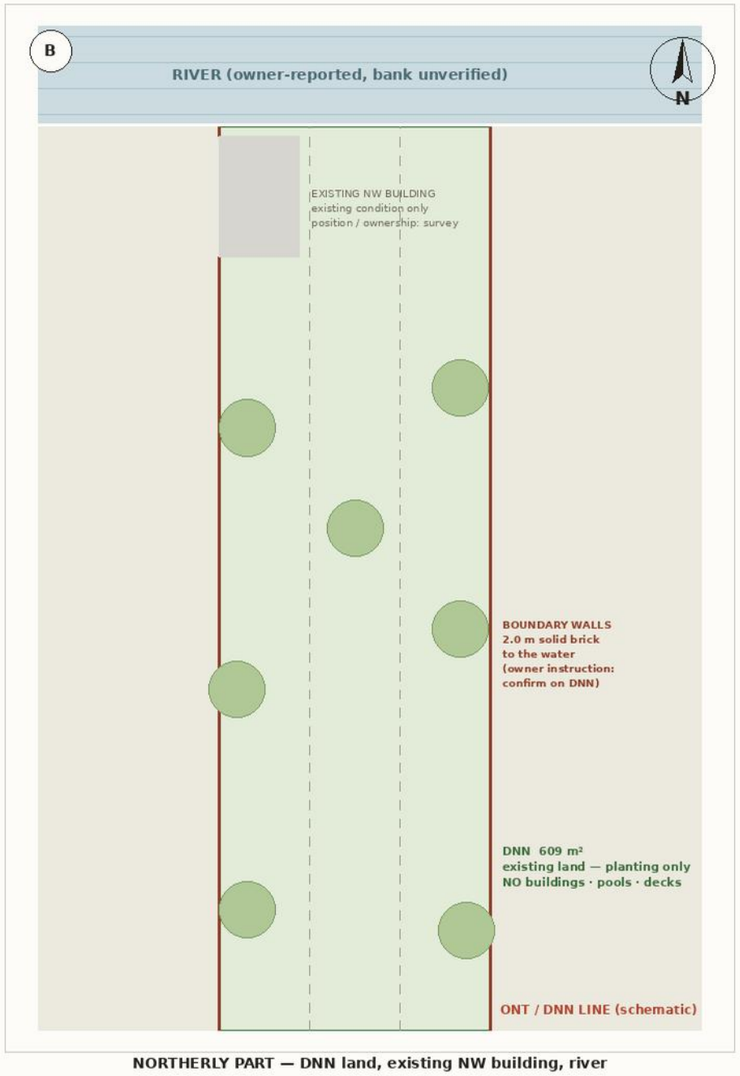
The house is fitted to the 13.5 m frontage at its brochure width of 10.4 m. That leaves a 2.8 m garden strip on the west, and the garage sits 0.3 m off the east party wall. The rear wall is at y 16.2 m, so the alfresco (4.3 × 2.8 m) stays inside the ONT band, ending at 19.0 m against the 19.73 m limit. Nothing is built on the DNN land except the boundary walls. The garage door is set 3.9 m behind the 2.0 m front wall, which is enough for the gate leaves and a short apron, but not to park in front of the garage. The plan is adapted from a commercial volume-builder design for study only. Building it would need the design owner's licence or a new design.

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

SITE PLAN · 1:200 at A3 · shown in two parts so the narrow 13.5 m plot stays readable



SOUTHERLY PART — ONT band, house, road



NORTHERLY PART — DNN land, existing NW building, river

NOTES

Schematic site from owner information only: three side-by-side 4.5 m titles, about 13.5 × 65 m. No survey. Corners, levels, bank edge and road reserve are unverified.

Structures are shown ONLY inside the ONT road-side band (3 × 88.8 = 266.4 m²). The 609 m² DNN band is existing land, shown as planting only.

The title sketch shows 6.5 m to the road centreline. It is NOT treated as a verified setback.

Assumed: a 3-storey house on each side built to the property line to about the same 20 m depth. So the east wall is blank (party wall) and the west wall has high-level windows only.

Side boundary walls run the full length to the water on both property lines (owner instruction). This is the only structure on DNN land. Confirm it is permitted.

The existing north-west building is an existing condition only. Its position and ownership are to be surveyed. It is not altered.

Subject to legal merger of the three ONT sections, setbacks, river corridor and permits.

KEY

- ONT — buildable band
- DNN — land / planting only
- service passage
- existing NW building
- assumed 3-storey neighbours
- boundary walls
- house footprint
- assumed / context

0 10 20 30 40 50 metres (assumed — no survey)

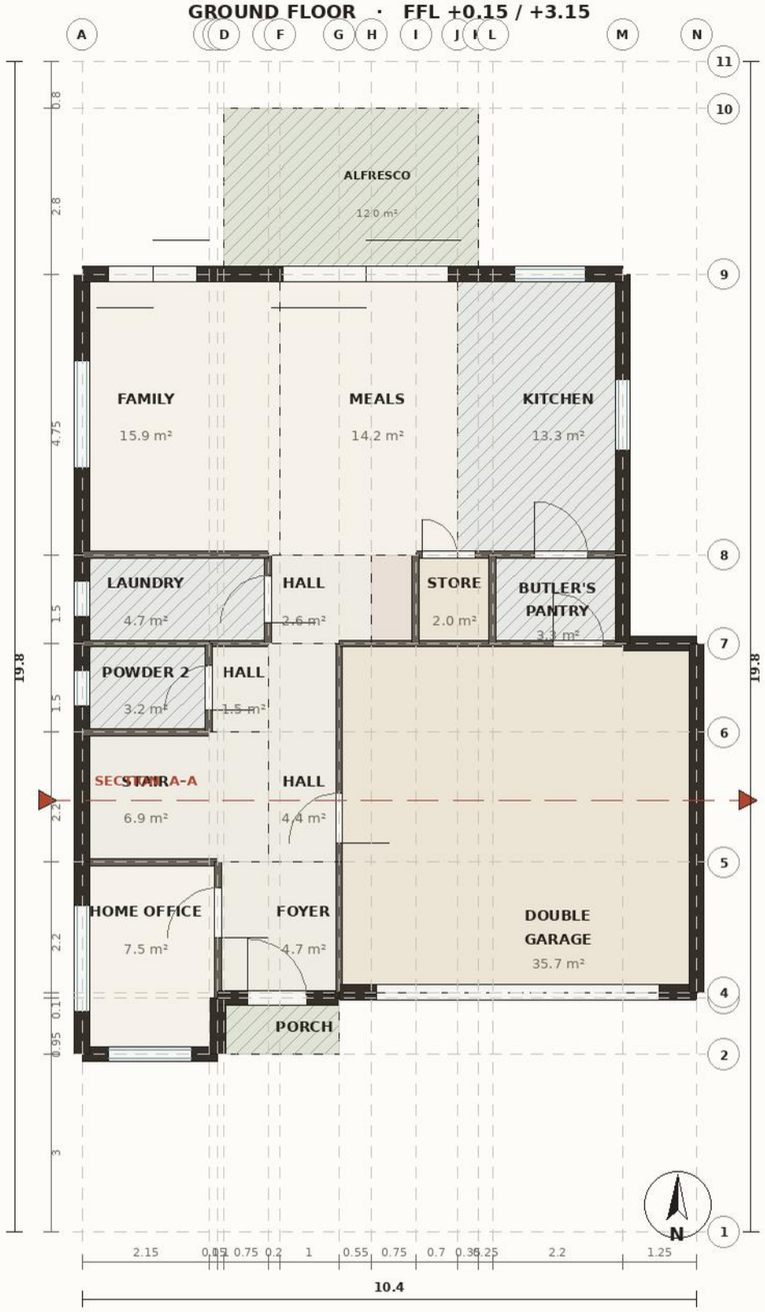
CONCEPT 3 — FOUR-BED GARAGE HOUSE

Two storeys · double garage · home office · open kitchen, meals and family to a rear alfresco · four bedrooms + rumpus

SCALE	SHEET	STATUS
1:200 at A3	**C3-01** site plan	**FOR REVIEW** *not for construction*

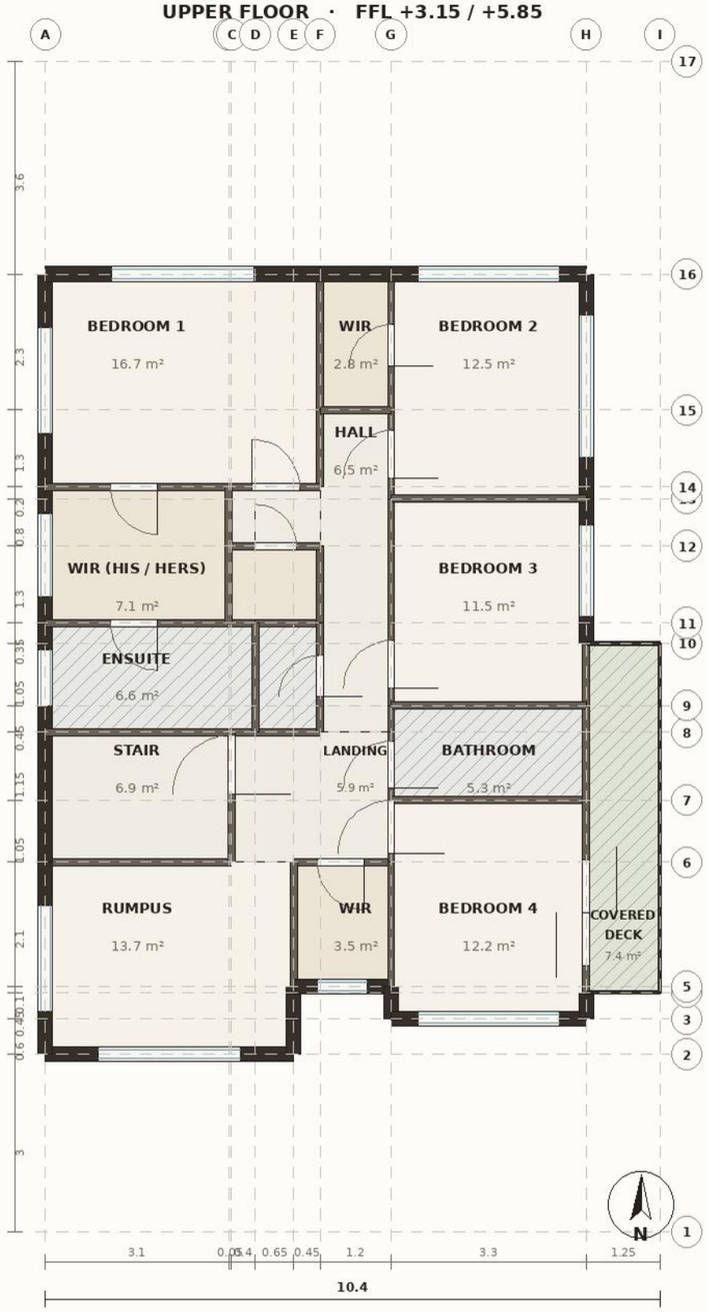
enclosed 123.3 m² · level 1 of 2

GROUND FLOOR · FFL +0.15 / +3.15



enclosed 129.9 m² · level 2 of 2

UPPER FLOOR · FFL +3.15 / +5.85



0 10 20 30 40 50 metres (assumed — no survey)

KEY

- HABITABLE
- SERVICE / WET
- CIRCULATION
- STORAGE
- SPECIAL
- OUTDOOR
- PLANT / VOID

NOTES

CONCEPT ONLY — NOT FOR CONSTRUCTION. Dimensions are taken from owner-supplied title information; no survey. Subject to survey, legal merger, setbacks and permits.

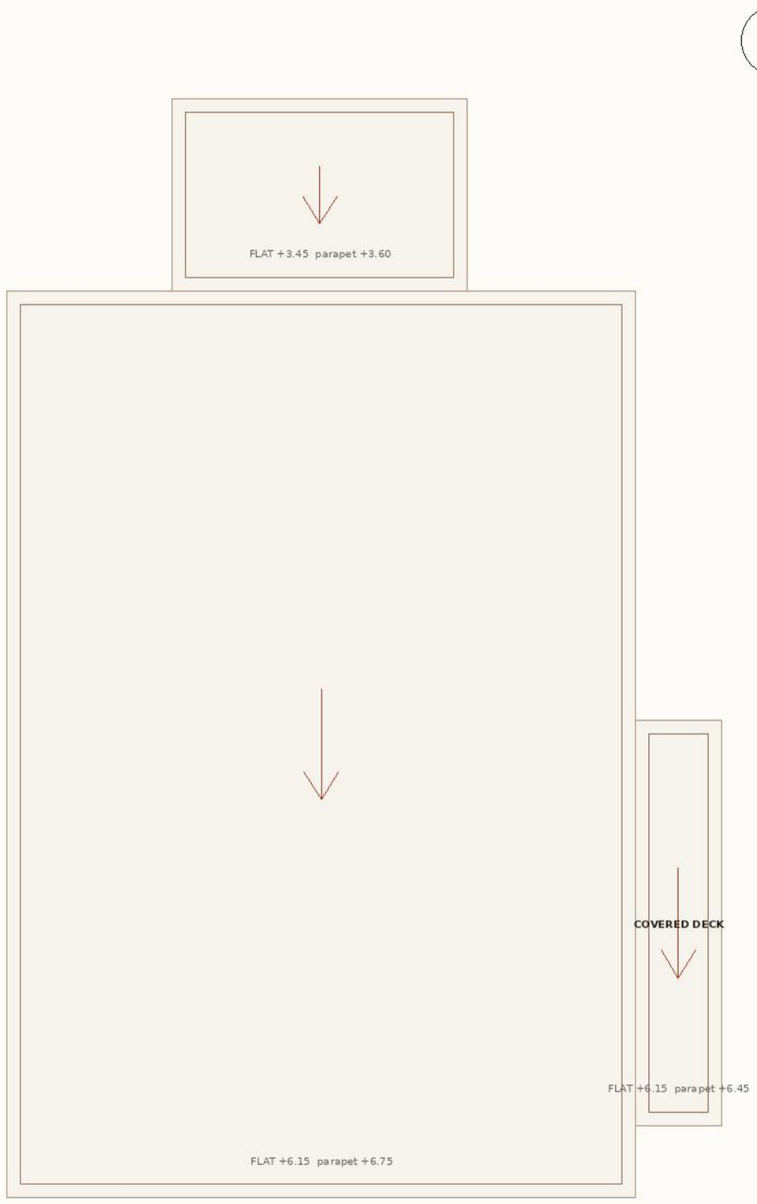
Areas are measured from the room rectangles; the balcony and light court are excluded from enclosed area.

CONCEPT 3 — FOUR-BED GARAGE HOUSE

Two storeys · double garage · home office · open kitchen, meals and family to a rear alfresco · four bedrooms + rumpus

SCALE	SHEET	STATUS
1:100 at A3	C3-02	FOR REVIEW
	floor plans 1–2	*not for construction*

ROOF PLAN · looking down



0 10 20 30 40 50 metres (assumed — no survey)

KEY

- roof covering
- fall arrow
- parapet
- open slot below

ROOF KEY

- membrane flat roofs behind parapets — deck +6.15, parapet +6.75
- fall arrow (to box gutter / outlet)
- parapet inner line
- outline of the level below (hidden line)

NOTES

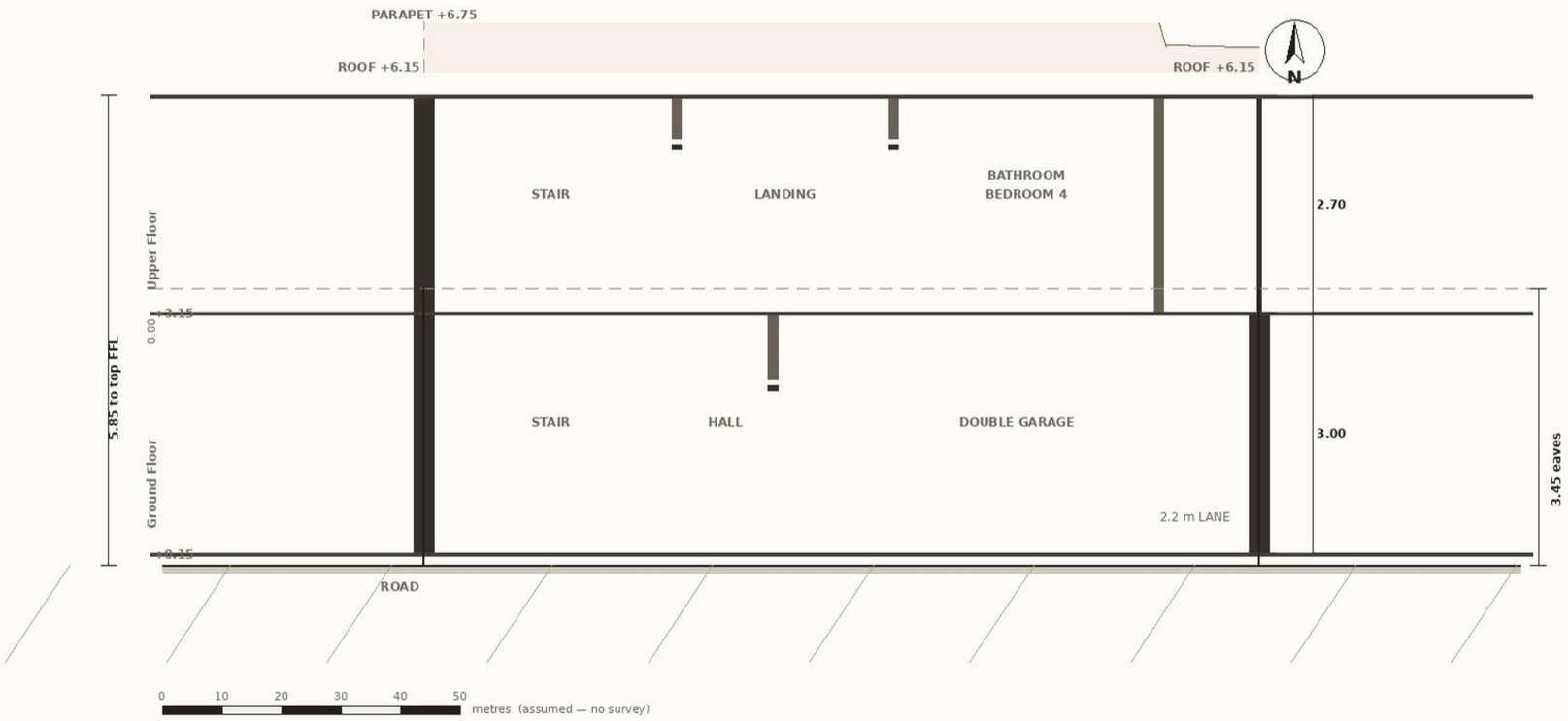
Flat roofs: membrane deck +6.15 m, parapets up to +6.75 m above datum, no overhang. Falls and outlets to be set at design development.
Roof outlines come from the same rectangles as the plans and the 3D model.
Rainwater: box gutters behind the parapets to downpipes inside the site; nothing may discharge onto the neighbours.

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SCALE	SHEET	STATUS
1:100 at A3	**C3-03** roof plan	**FOR REVIEW** *not for construction*

SECTION A–A · looking north at y = 7.30 m (assumed cut)



existing ground level ±0.00 — all levels assumed

KEY

- cut external wall, 0.30
- cut internal wall, 0.15
- glazing in the cut
- roof build-up in section

NOTES

Datum 0.00 = existing road and ground level. Finished ground floor +0.15 m, first +3.15 m, eaves +3.45 m, ridge +6.75 m.

Slabs, walls and levels come from the same rectangles as the plans, so the section and the plans can never disagree.

The 2.2 m rain-garden slot is open to the sky on every level: draw it as a full-height void when the structure is detailed.

The cut is at y = 7.30 m. Move it on the plans to suit a later survey.

CONCEPT 3 — FOUR-BED GARAGE HOUSE

Two storeys · double garage · home office · open kitchen, meals and family to a rear alfresco · four bedrooms + rumpus

SCALE	SHEET	STATUS
1:100 at A3	**C3-04** section A–A	**FOR REVIEW** *not for construction*

AREA SCHEDULE · net internal, measured from the plan rectangles

Room	Type	Area m²	Cum. m²
Ground Floor · FFL +0.15			
Family	Habitable	15.9	15.9
Meals	Habitable	14.2	30.2
Home Office	Habitable	7.5	37.6
Alfresco	Outdoor	12.0	49.7
Foyer	Circulation	4.7	54.4
Double Garage	Storage	35.7	90.1
Hall	Circulation	4.4	94.5
Hall	Circulation	2.6	97.2
Hall	Circulation	1.5	98.7
IT Nook	Special	1.1	99.8
Kitchen	Service / Wet	13.3	113.1
Laundry	Service / Wet	4.7	117.8
Butler's Pantry	Service / Wet	3.3	121.1
Porch	Outdoor	1.9	123.0
Powder 2	Service / Wet	3.2	126.3
Stair	Circulation	6.9	133.2
Store	Storage	2.0	135.2
Level total, enclosed		**123.3**	
Upper Floor · FFL +3.15			
Bedroom 1	Habitable	16.7	16.7
Bedroom 2	Habitable	12.5	29.3
Bedroom 3	Habitable	11.5	40.8
Bedroom 4	Habitable	12.2	53.0
Rumpus	Habitable	13.7	66.7
Bathroom	Service / Wet	5.3	72.0
Covered Deck	Outdoor	7.4	79.3
Ensuite	Service / Wet	6.6	85.9
Hall	Circulation	6.5	92.5
Hall	Circulation	1.5	94.0
Landing	Circulation	5.9	99.9
Powder 1	Service / Wet	2.0	102.0
Stair	Circulation	6.9	108.9
Linen	Storage	2.0	110.9
WIR (his / hers)	Storage	7.1	118.1
WIR	Storage	2.8	120.8
WIR	Storage	3.5	124.3
Level total, enclosed		**129.9**	

AREA SUMMARY	Area
Level 1 — Ground Floor (enclosed)	123.3 m²
Level 2 — Upper Floor (enclosed)	129.9 m²
Total enclosed floor area	**253.2 m²**

ORDER OF COST — TURNKEY, NORTHERN VIET NAM

INDICATIVE BUILD COST	M VND	+10% prov.
Basic build — 4.9 M VND/m²	1 241	1 365
Mid build — 6.5 M VND/m²	1 646	1 810
High build — 8.5 M VND/m²	2 152	2 367

NOTES

Areas are net internal floor areas computed from the room rectangles. Terraces, loggias, the rain-garden slot, the plant deck and roof gardens are excluded from the totals but shown on the plans.

The build-cost range is a rule of thumb from published Vietnamese cost guides for northern Viet Nam, not a quotation. It excludes site works, boundary treatment and furniture.

Confirm with a local quantity surveyor and at least two builders before committing to a budget.

NOTES

No habitable room here is below 2.7 m clear. The ground floor is a complete one-level home — bedroom, bathroom, living, dining and kitchen all at +0.15 — so the family can live on one level for as long as they want to.

Order-of-cost at 4.9–8.5 M VND/m² turnkey (northern Viet Nam) is in the table on the left. It is a rule of thumb, not a quotation.

ROOM TYPE KEY

- HABITABLE
- SERVICE / WET
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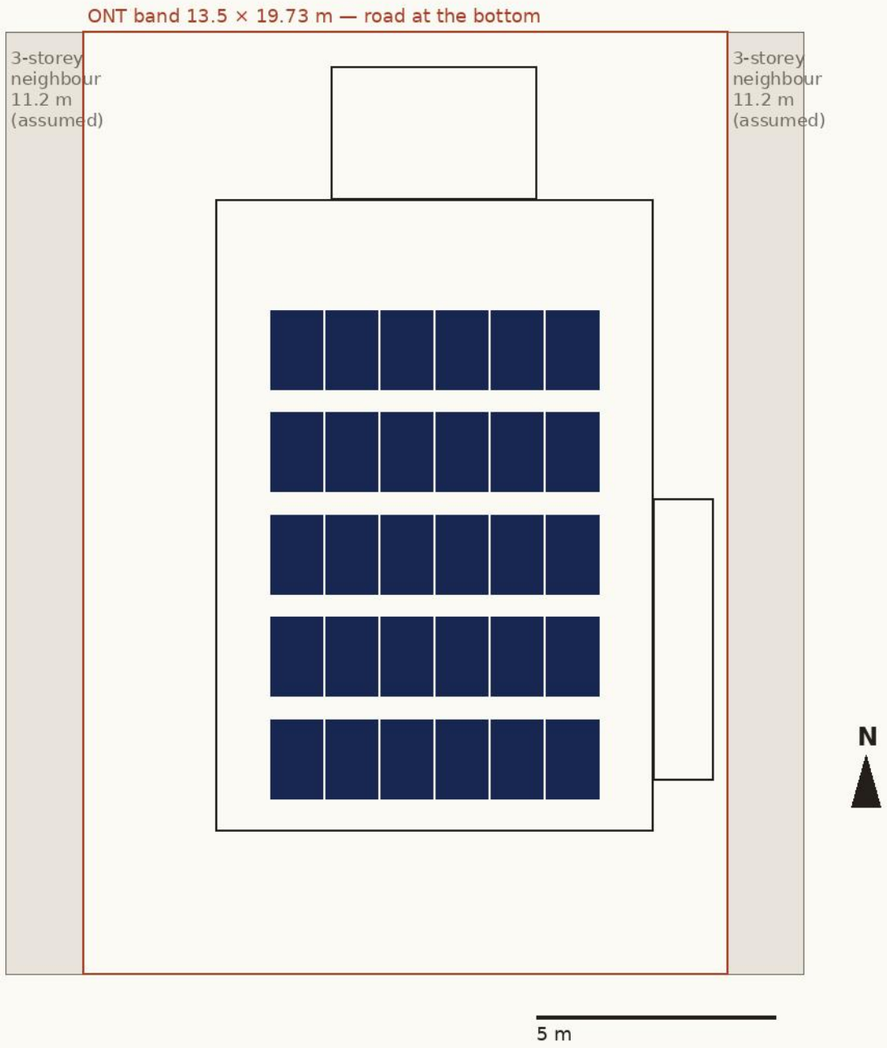
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SCALE	SHEET	STATUS
as noted	**C3-05** area schedule	**FOR REVIEW** *not for construction*

Rooftop solar — energy per season

30 × 440 Wp panels = 13.2 kWp on the roof · latitude 21°N (rounded) · PVGIS 5.3 TMY (ERA5), typical year · shaded by the assumed 3-storey neighbours



Recommended panels

Fit but not worth it: number = kWh per kWp a year (an unshaded ideal panel gives 1281)

Roof surface	Tilt / faces	Fit / used	kWh/kWp	no neighbours
Main roof flat roof	10° S	30 / 30	1068	1265

ENERGY FROM THE RECOMMENDED ARRAY

Season	Months	kWh	kWh / day	Share
Winter	Dec–Feb	2,691	29.9	19 %
Spring	Mar–May	3,829	41.6	27 %
Summer	Jun–Aug	3,940	42.8	28 %
Autumn	Sep–Nov	3,638	40.0	26 %
Year		**14,098**	**38.6**	

Without the assumed neighbours the same panels would give 16,702 kWh a year (+18 %). All 30 panels that physically fit would give 14,098 kWh, but the extra ones are on shaded or east/west/north slopes.



IS IT WORTH IT? (indicative)

	Recommended	Best-value size
Panels / kWp	30 / 13.2	6 / 2.6
Installed cost	185 M VND	37 M VND
Bill saving, first year	9 M VND	6 M VND
Output used in the house	18 %	60 %
Share of the bill-kWh covered	36 %	26 %
Simple payback	over 25 years	6 years
Net over 25 years	-17 M VND	106 M VND

Assumes: an example air-conditioned family using about 6,890 kWh a year (bill 21 M VND a year), EVN household tiers from 1 Nov 2026 + 8 % VAT, 14 M VND per kWp installed, nothing paid for power sent to the grid (the buy-back rate is not yet published), 0.5 %/yr panel ageing, 1 %/yr upkeep. Surplus power is the limit: the house uses most of its power in the evening. Try other numbers at ali3b.github.io/home-design/solar.html.

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Batteries and living off-grid

Hour-by-hour over a typical year · battery 5 M VND per usable kWh incl. hybrid inverter, replaced once (year 13, 60 %) · nothing paid for surplus power

WITHOUT AIR-CON — example household using about 3,600 kWh a year (bill 9 M VND a year)

Panels	Battery	Cost	From sun: W Sp Su Au	Year	No-grid days	Saving / yr	Payback	Net over 25 years
6 (2.6 kWp)	none	37 M VND	26 30 31 28	29 %	0	3 M VND	13 yrs	30 M VND
6 (2.6 kWp)	**5 kWh**	**62 M VND**	54 68 68 66	**64 %**	**0**	**6 M VND**	**10 yrs**	**66 M VND ← best value**
6 (2.6 kWp)	10 kWh	87 M VND	59 82 83 79	76 %	69	7 M VND	18 yrs	46 M VND
6 (2.6 kWp)	15 kWh	112 M VND	59 83 84 79	76 %	75	7 M VND	24 yrs	7 M VND
30 (13.2 kWp)	none	185 M VND	35 42 45 38	40 %	0	4 M VND	> 25 yrs	-129 M VND
30 (13.2 kWp)	5 kWh	210 M VND	77 86 89 82	83 %	0	8 M VND	> 25 yrs	-84 M VND
30 (13.2 kWp)	10 kWh	235 M VND	98 100 100 100	100 %	358	9 M VND	> 25 yrs	-95 M VND
30 (13.2 kWp)	15 kWh	260 M VND	99 100 100 100	100 %	360	9 M VND	> 25 yrs	-135 M VND

WITH AIR-CON — example household using about 6,890 kWh a year (bill 21 M VND a year)

Panels	Battery	Cost	From sun: W Sp Su Au	Year	No-grid days	Saving / yr	Payback	Net over 25 years
6 (2.6 kWp)	none	37 M VND	25 28 25 25	26 %	0	6 M VND	6 yrs	106 M VND
6 (2.6 kWp)	**5 kWh**	**62 M VND**	47 45 31 43	**39 %**	**0**	**10 M VND**	**7 yrs**	**140 M VND ← best value**
6 (2.6 kWp)	10 kWh	87 M VND	50 47 31 45	41 %	0	10 M VND	9 yrs	106 M VND
6 (2.6 kWp)	15 kWh	112 M VND	50 47 31 45	41 %	0	10 M VND	17 yrs	66 M VND
30 (13.2 kWp)	none	185 M VND	34 38 37 35	36 %	0	9 M VND	> 25 yrs	-17 M VND
30 (13.2 kWp)	5 kWh	210 M VND	69 62 52 59	59 %	0	14 M VND	20 yrs	58 M VND
30 (13.2 kWp)	10 kWh	235 M VND	97 84 67 84	80 %	116	18 M VND	18 yrs	106 M VND
30 (13.2 kWp)	15 kWh	260 M VND	97 95 81 95	90 %	205	20 M VND	18 yrs	110 M VND

CAN THE HOUSE LIVE OFF-GRID?

Without air-con: YES in a typical year with 30 panels (13.2 kWp) and a 40 kWh battery, about 385 M VND. That costs 335 M VND more over 25 years than staying on the grid.

Nearly: 30 panels with 10 kWh covers all but 14 kWh a year (7 short days, the worst needing 3.1 kWh).

With air-con: NOT on this roof. The best tested (30 panels, 40 kWh battery, 385 M VND) runs 99 % of the year's power from the sun, but still needs 43 kWh from a generator or the grid, with up to 2 days in a row short.

Nearly: 30 panels with 40 kWh covers all but 43 kWh a year (9 short days, the worst needing 11.3 kWh).

A "typical year" leaves out the worst years. Northern winters bring weeks of drizzle and low cloud, so an off-grid house needs a margin: about 30 % more battery and panels, or a small generator for bad spells. Off-grid also wastes the summer surplus. Staying connected with a 5–10 kWh battery gets most of the benefit for far less money, and keeps the lights on in power cuts.

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Blender views

Cycles renders from the same data as the drawings and the walkthrough



Street view



From the garden (DNN land, planting only)

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Blender views

Cycles renders from the same data as the drawings and the walkthrough



From the river end, between the boundary walls



Foyer looking down the hall to the living area

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Blender views

Cycles renders from the same data as the drawings and the walkthrough



Family, meals and kitchen opening to the alfresco

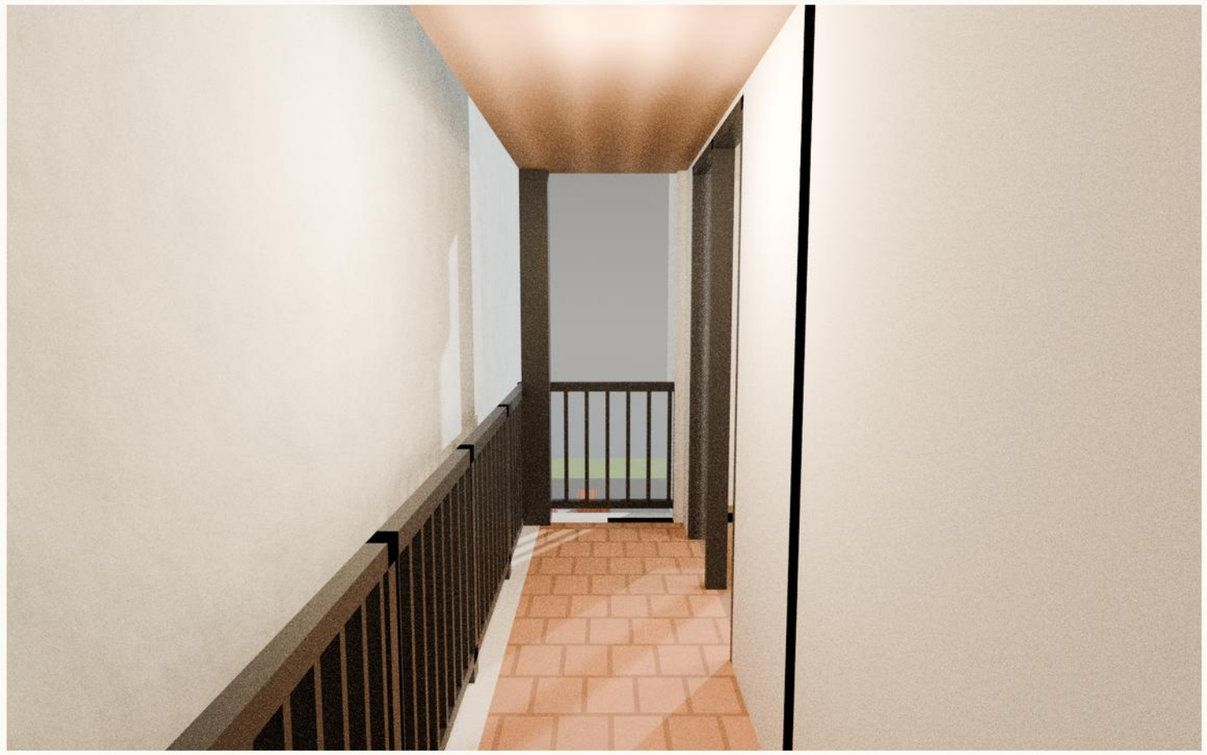


Aerial

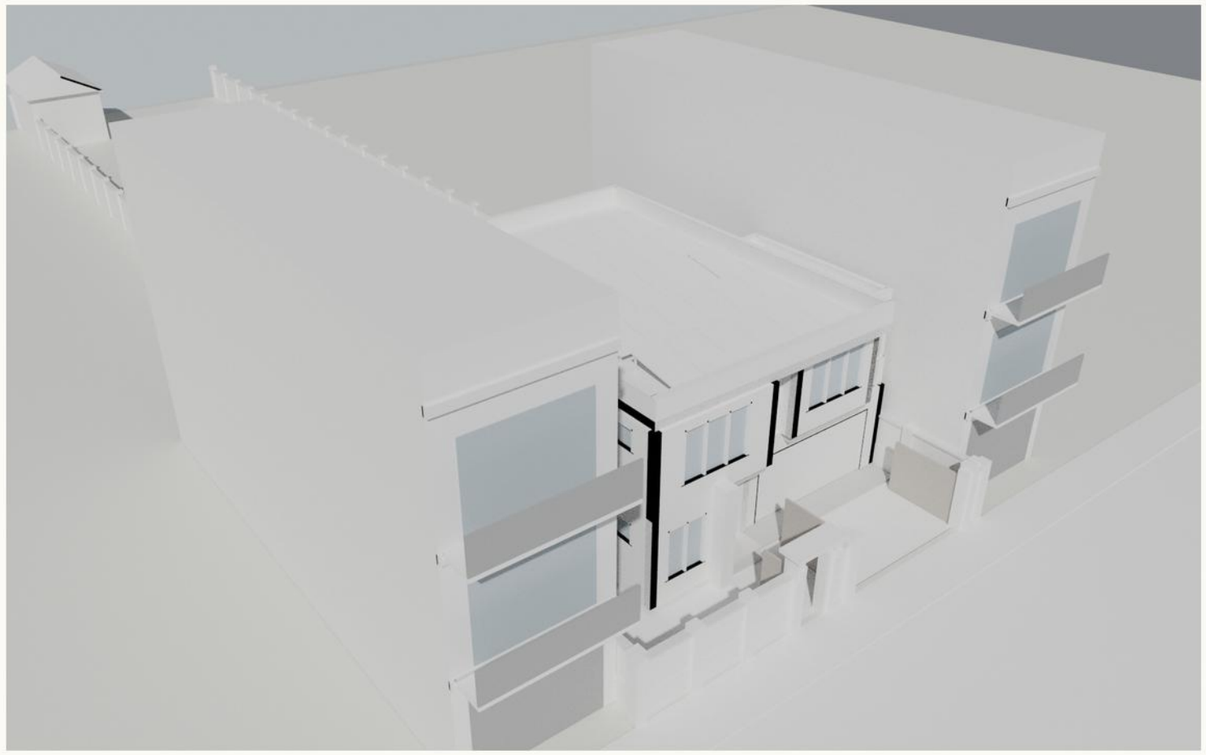
CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Blender views

Cycles renders from the same data as the drawings and the walkthrough



Covered deck on the garage roof



Clay massing

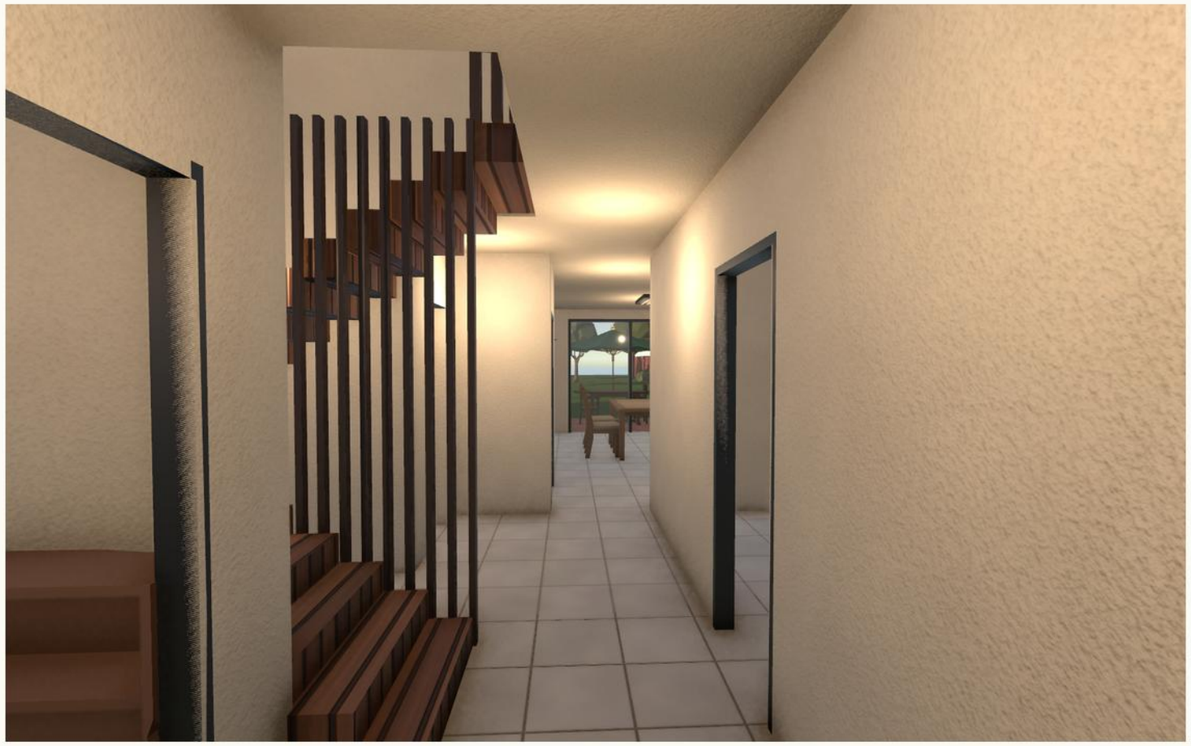
CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Walkthrough (Unity)

Interactive model: these are fixed viewpoints you can jump to with keys 1–9



Street



Foyer



Family



Kitchen

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Walkthrough (Unity)

Interactive model: these are fixed viewpoints you can jump to with keys 1–9



Garage



Rumpus



Bed1

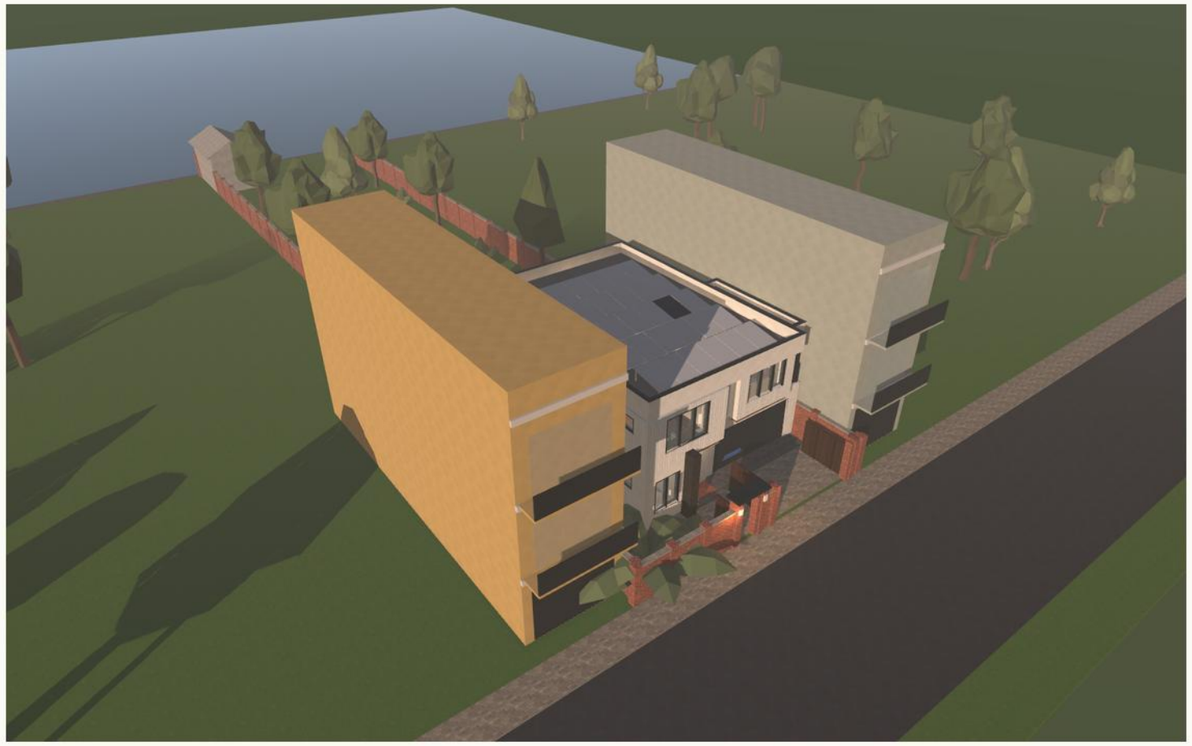


Deck

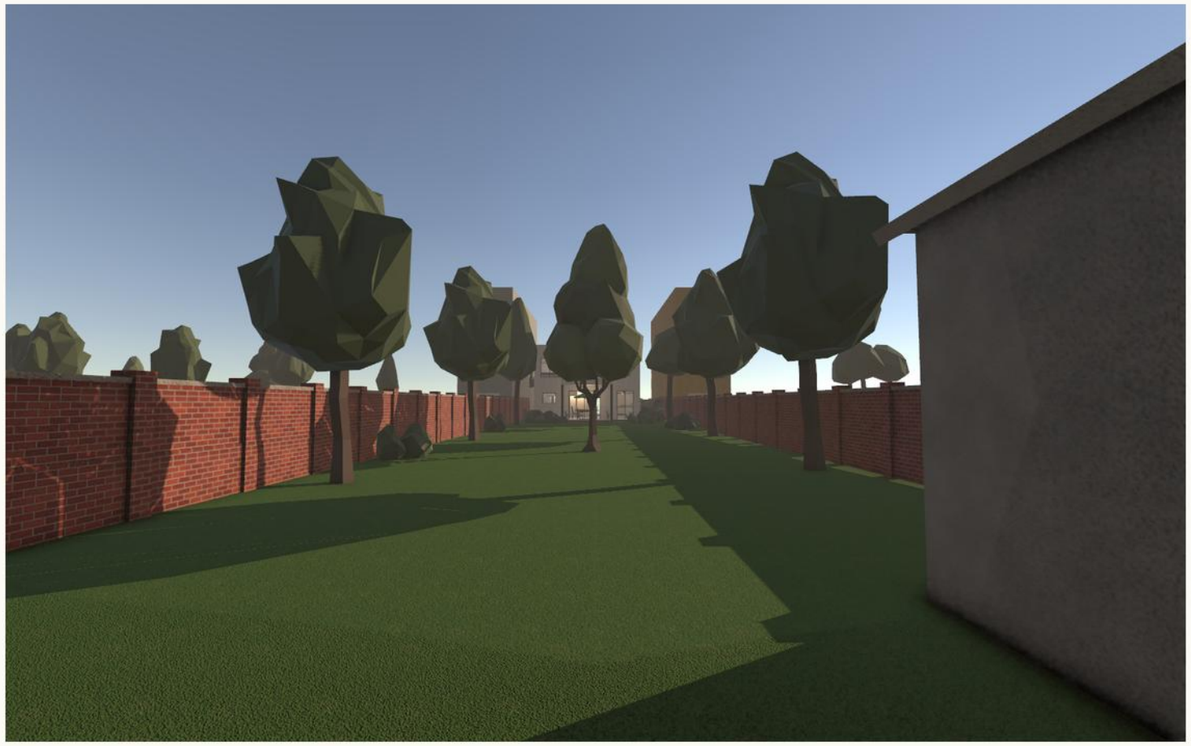
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Walkthrough (Unity)

Interactive model: these are fixed viewpoints you can jump to with keys 1–9



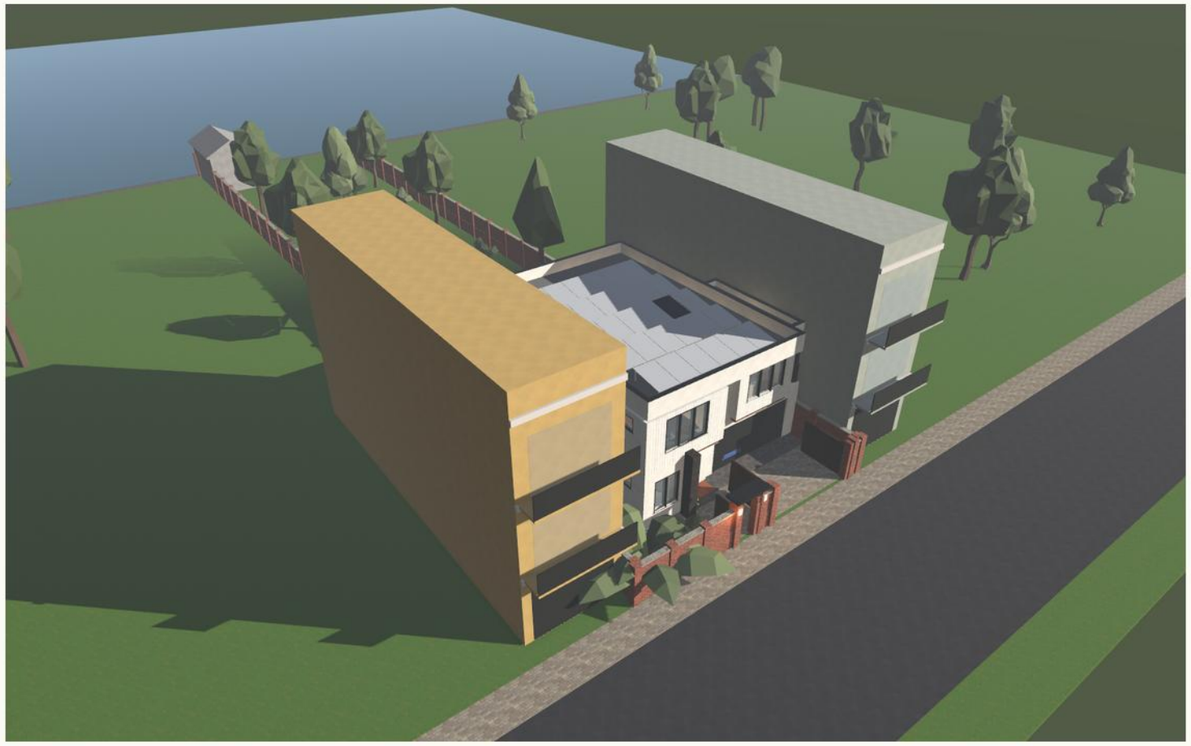
Aerial



Riverside



Dollhouse Ground



Sun study: winter solstice, 09:00 solar time

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Walkthrough (Unity)

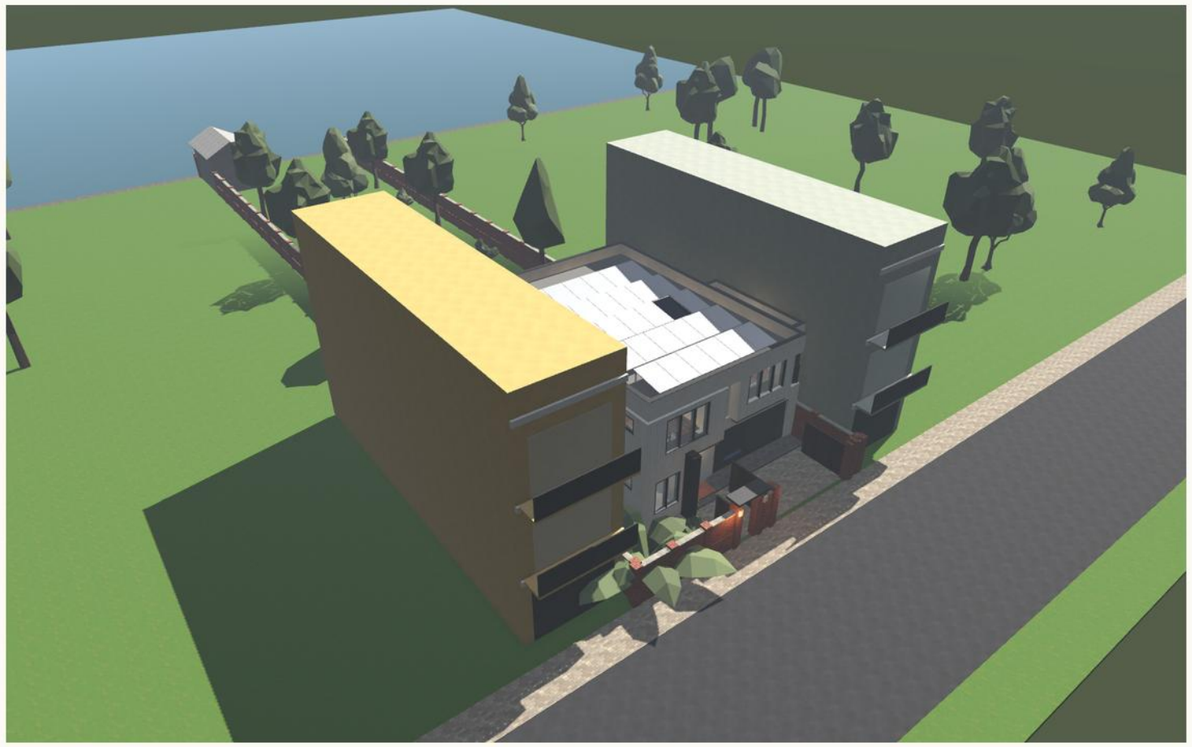
Interactive model: these are fixed viewpoints you can jump to with keys 1–9



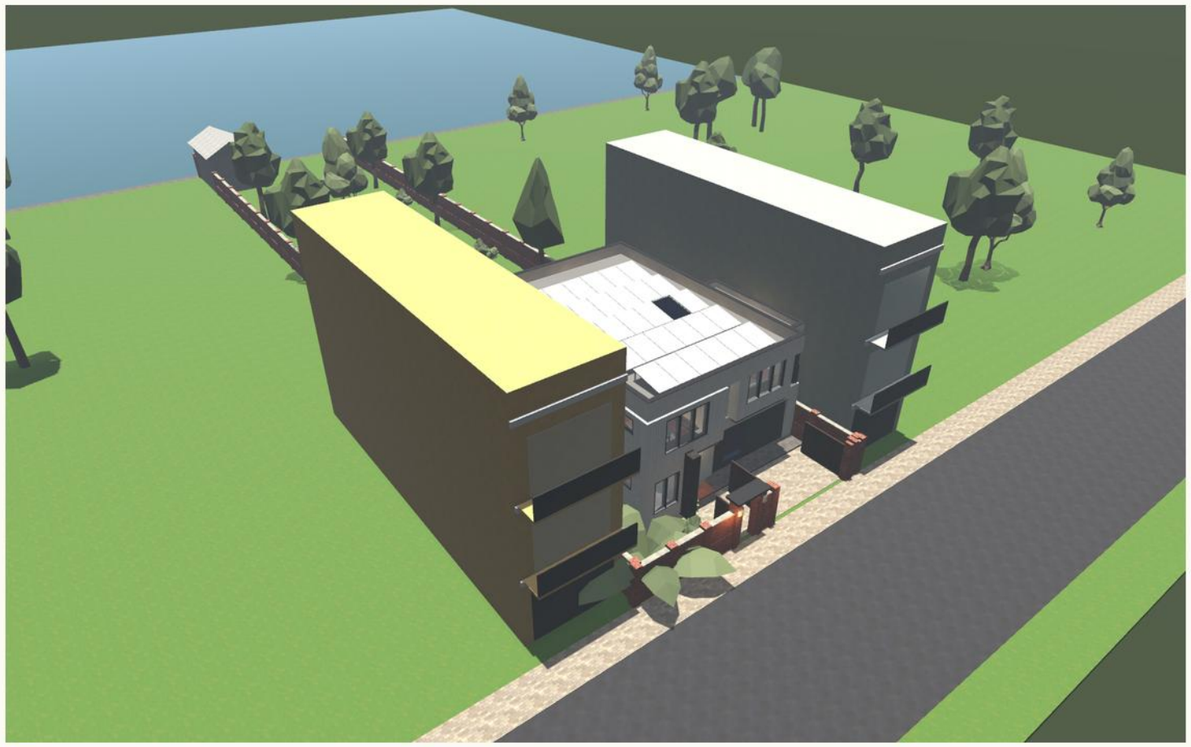
Sun study: winter solstice, 12:00 solar time



Sun study: winter solstice, 15:00 solar time



Sun study: summer solstice, 09:00 solar time

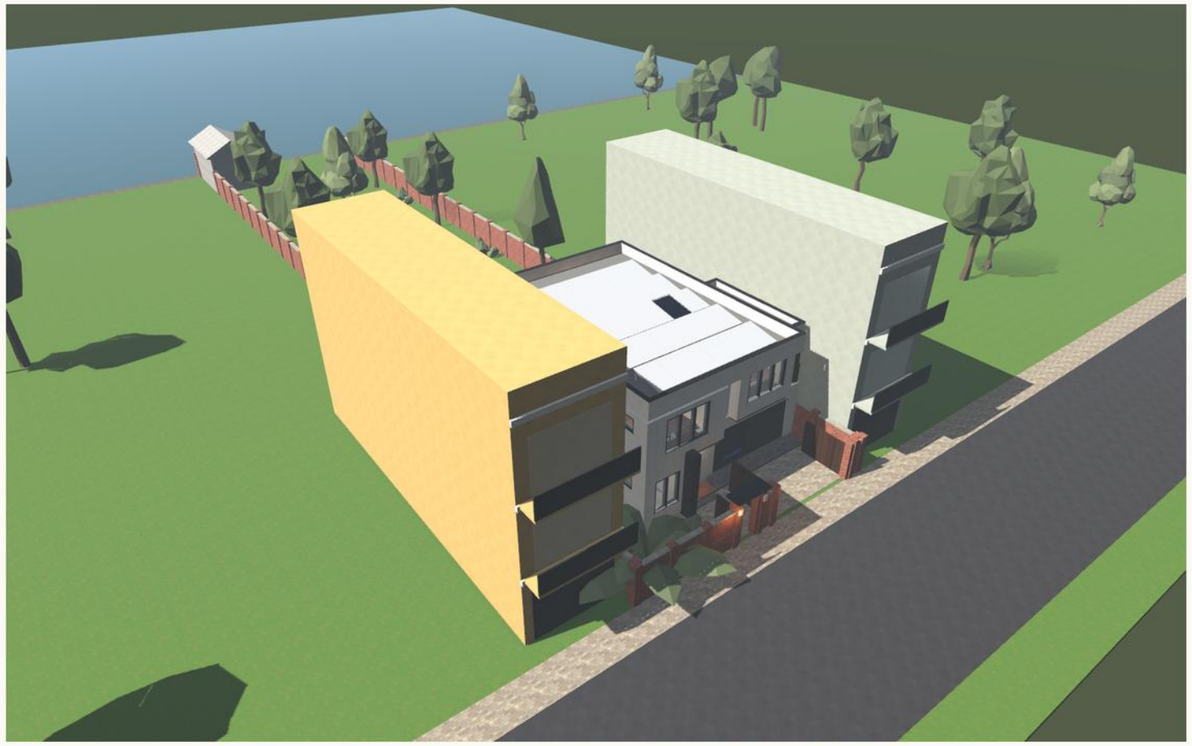


Sun study: summer solstice, 12:00 solar time

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Walkthrough (Unity)

Interactive model: these are fixed viewpoints you can jump to with keys 1–9



Sun study: summer solstice, 15:00 solar time

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